

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 5.2: Monte Carlo, Exploration, and Temporal-Difference Learning

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2026-07-08 01:00:15

Lecture 5 in One Map — Act 5.2

Act 5.1 ended with three departures on the table. This act executes #1 — replace the full-expectation backup, which needs a known model (P, R), with a sample backup: Monte Carlo, TD, Q-learning — and pays sampling's price, #3: exploration.



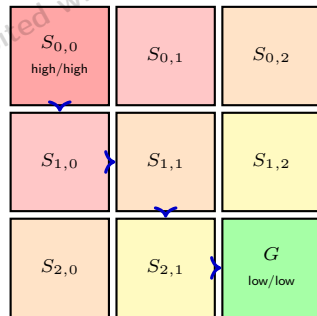
The internal flow of this act: a concrete economic gridworld → **Monte Carlo** (learn from whole episodes) → the **exploration–exploitation** problem → **temporal-difference** and **Q-learning** (update every step). One backbone underneath all of them: *Generalized Policy Iteration*.

A Motivating Example

A small economic gridworld to make “learning from experience” concrete

A 3×3 Economic Grid

- A **policymaker** navigates macroeconomic conditions (inflation × unemployment).
- Goal: reach and maintain the optimal state (low inflation, low unemployment).
- **Top-left** ($S_{0,0}$): High inflation & unemployment (worst).
- **Bottom-right** ($G \equiv S_{2,2}$): Low inflation & low unemployment (**goal**).
- Episode ends upon reaching the goal.
- Optimal trajectory (arrows) shows one greedy path; many equivalent paths exist.



Macro analogue of textbook gridworld.

Actions and Rewards on the Grid

Actions (Policy Interventions):

- **Up**: Contractionary policy (reduces inflation, may raise unemployment).
- **Down**: Expansionary policy (reduces unemployment, may raise inflation).
- **Left/Right**: Fiscal or structural interventions.

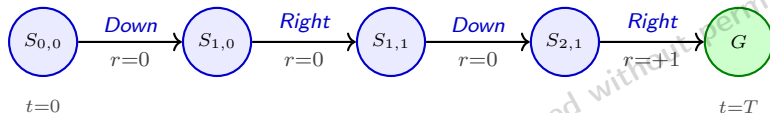
Reward Structure:

$$R(s, a) = \begin{cases} +1 & \text{if next state} = \text{Goal state } G, \\ 0 & \text{otherwise.} \end{cases}$$

MDP on the grid: $\langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$ where $\mathcal{S} = \{S_{i,j}\}$, $\mathcal{A} = \{\text{Up, Down, Left, Right}\}$.

Illustrative Episode

Example rollout ($r_t = 0$ until goal; $r_T = +1$):



- Cumulative return $G_0 = \sum_{t=0}^{T-1} \gamma^t r_{t+1} = \gamma^{T-1}$.
- Update $Q(s, a)$ by averaging returns over many such episodes.
- Repeated simulation $\rightarrow$ stable policy guiding the economy toward G .

How do we compute updates? Three families:

1. **Monte Carlo** — wait for the full episode, then average.
2. **Temporal Difference** — update after every single step.
3. **Actor-Critic** — combine policy learning with value estimation.

Monte Carlo Methods

Learn from complete simulated episodes — no model required

Model-Based DP vs. Monte Carlo

Model-Based Policy Iteration (Classical DP):

1. **Policy evaluation:** Solve the Bellman equation:

$$V^\pi(s) = \sum_a \pi(a|s) [R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s')].$$

2. **Policy improvement:** $\pi'(s) = \arg \max_a Q^\pi(s, a)$.

Requirement: Must know $P(s'|s, a)$ and $R(s, a)$.

Monte Carlo (MC):

- Learn through **simulated episodes** without requiring explicit transition models.
- Estimate $Q(s, a)$ from **observed returns** instead of computing expectations.
- Like running a randomized policy experiment many times and averaging the outcomes.

Basic Monte Carlo: Key Idea

Simulate full episodes, then average the returns.

- Simulate multiple **full episodes** from each (s, a) .
- Collect total discounted return:

$$G_t = \sum_{k=0}^{T-1} \gamma^k r_{t+k+1}.$$

- Update rule:

$$Q(s, a) \leftarrow \frac{1}{N} \sum_{i=1}^N G_0^{(i)},$$

where N = number of episodes, $G_0^{(i)}$ = return from episode i .

Why full episodes?

- Captures **long-run consequences** of actions.
- Avoids dependence on explicit transition models.
- By the Law of Large Numbers, estimates converge.

Monte Carlo Applied to the Growth Model

Environment: State (k_t, z_t) , action c_t , reward $u(c_t)$, transition $k_{t+1} = F(k_t, z_t) - c_t$.

MC Episode:

1. Initialize k_0, z_0 .
2. For $t = 0, 1, \dots, T - 1$:
 - Choose c_t from current (or exploratory) policy.
 - Observe $r_t = u(c_t)$, transition to (k_{t+1}, z_{t+1}) .
3. Compute $G_0 = \sum_{t=0}^{T-1} \gamma^t r_t$.

Key difference from VFI:

- VFI: evaluate V at **every grid point**, using known $P(z'|z)$.
- MC: evaluate V along **sampled trajectories**, no need for explicit P .

Challenges of Basic Monte Carlo

Full-Episode Dependence:

- Updates only after **entire episodes** are completed.
- Inefficient for long-horizon or continuing (non-terminating) problems.

Exploration Dilemma:

- Infrequent visits to rare but important states (e.g., financial crises).
- Theory requires every (s, a) visited infinitely often.

High Variance:

- Returns from full episodes vary significantly.
- Slow convergence $\Rightarrow$ many episodes needed.

These motivate **exploring starts** / **subepisodes** (coverage), **explicit exploration** (ϵ -greedy, Boltzmann, UCB), and **incremental TD** updates.

Subepisodes and Exploring Starts

Subepisodes (First-Visit vs. Every-Visit MC):

- **First-Visit MC:** Update $Q(s, a)$ only on the **first** occurrence per episode.
- **Every-Visit MC:** Update $Q(s, a)$ on **every** occurrence.
- Both speed up convergence by leveraging all data within an episode.

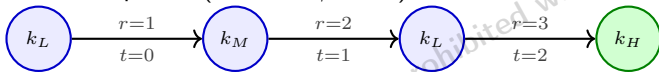
Exploring Starts:

- Each episode starts from a randomly chosen (s_0, a_0) with nonzero probability.
- Ensures every (s, a) pair receives enough visits.
- **Problem:** often unrealistic in practice (can't arbitrarily reset the economy).

⇒ We need a more practical exploration mechanism: ϵ -**greedy**.

First-Visit vs. Every-Visit: a Worked Trajectory

One simulated episode (discount $\gamma = 0.9$), with state k_L visited **twice**:



- Return from $t=0$: $G_0 = 1 + 0.9(2) + 0.9^2(3) = 5.23$. From $t=2$: $G_2 = 3$.
- **First-visit MC**: use G_0 only $\Rightarrow \hat{V}(k_L) = 5.23$ (one clean, independent sample).
- **Every-visit MC**: use *both* $\Rightarrow \hat{V}(k_L) = \frac{5.23+3}{2} = 4.12$ (more data, but correlated samples).

Trade-off: every-visit squeezes more from each episode; first-visit keeps samples independent.

Generalized Policy Iteration (GPI): the RL Backbone

Two steps, alternated until stable:

1. **Evaluate** — estimate $\hat{Q}^\pi$ (MC: average sampled returns; DP: full backup).
2. **Improve** — act greedily:
 $\pi'(s) = \arg \max_a \hat{Q}^\pi(s, a).$

MC control = MC prediction inside GPI.



$\rightarrow (Q^*, \pi^*)$

One backbone, every algorithm. DP, MC, TD, SARSA, Q-learning, and (in T3) actor-critic are *all* GPI — they differ only in *how evaluation approximates the expectation* and *how improvement explores*.

GPI in Action: a Two-State “Ping-Pong”

Toy growth model: capital $k \in \{k_L, k_H\}$; action = next capital k' . Current estimate $\hat{Q}$:

$\hat{Q}(k, k')$	$k' = k_L$	$k' = k_H$
$k = k_L$	3	5
$k = k_H$	8	6

- **Improve** (greedy): $\pi(k_L)=k_H$ ($5>3$), $\pi(k_H)=k_L$ ($8>6$). *Low capital* $\rightarrow$ *invest*, *high capital* $\rightarrow$ *consume* — the sensible rule.
- **Re-evaluate** $\hat{Q}$ under the new π , then improve again — “ping-pong” until $(\hat{Q}, \pi)$ stops moving: a fixed point $\approx (Q^*, \pi^*)$.

With only *sampled* $\hat{Q}$, a greedy step can occasionally *worsen* the policy (noise fools the $\arg\max$). Monotone improvement is guaranteed only with *exact* Q — hence the need for enough **exploration** (next).

Exploration vs. Exploitation

To learn a good policy you must sometimes act sub-optimally on purpose

The Exploration–Exploitation Tradeoff

Key Challenge in RL:

- **Exploitation:** Choose the best-known action to maximize *immediate* reward.
- **Exploration:** Try actions that may seem suboptimal now but could yield better *long-term* rewards.

Economic Analogies:

- **Dynamic Pricing:** Use the current best-known price, or test new prices?
- **Portfolio Management:** Stick to blue-chip stocks, or diversify into unknowns?
- **R&D Investment:** Exploit existing technology, or explore new innovations?
- **Monetary Policy:** Follow the Taylor rule, or experiment with novel instruments?

ϵ -Greedy Exploration

- With probability $1 - \epsilon$: select the **best-known action** (greedy).
- With probability ϵ : select an action **at random** (explore).

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}(s)|}, & \text{if } a = \arg \max_{a'} Q(s, a') \\ \frac{\epsilon}{|\mathcal{A}(s)|}, & \text{otherwise.} \end{cases}$$

- Ensures all actions have **nonzero probability** $\Rightarrow$ every (s, a) visited eventually.
- Simple, practical, widely used. ●
- Limitation: explores **uniformly** among non-greedy actions (doesn't favor promising ones).
- **GLIE** (convergence): decay $\epsilon_n \rightarrow 0$ slowly (e.g. $\epsilon_n = 1/n$) so every (s, a) is still visited infinitely often — then ϵ -greedy GPI converges to π^* .

Boltzmann (Softmax) Exploration

Idea: Actions chosen **probabilistically**, weighted by estimated value.

$$\Pr(a \mid s) = \frac{\exp(Q(s, a)/\tau)}{\sum_{a'} \exp(Q(s, a')/\tau)},$$

where $\tau > 0$ is the **temperature**:

- τ **high** $\Rightarrow$ near-uniform exploration.
- τ **low** $\Rightarrow$ near-greedy selection.

Connection to LLMs: Large Language Models use the same “temperature” parameter to balance creativity (exploration) and consistency (exploitation) in text generation.

Economic example: Online advertising allocates budget to ads $\propto \exp(\text{estimated ROI}/\tau)$.

ϵ -Greedy vs. Boltzmann: a Numerical Look

Three actions with current values $\hat{Q} = (10, 9, 1)$:

	a_1 ($Q=10$)	a_2 ($Q=9$)	a_3 ($Q=1$)
ϵ -greedy ($\epsilon=0.1$)	0.93	0.03	0.03
Boltzmann ($\tau=2$)	0.62	0.37	0.01

- ϵ -greedy spreads exploration **uniformly**: the near-optimal a_2 and the clearly-bad a_3 get the *same* probability.
- Boltzmann respects the **value ordering**: it explores the promising a_2 far more than the inferior a_3 .

The same softmax/temperature structure economists know from discrete-choice (logit) models — and from LLM decoding.

Upper Confidence Bounds (UCB)

Key idea: Give an **exploration bonus** to less-trying actions.

$$a^* = \arg \max_a \left[\hat{Q}(a) + \alpha \sqrt{\frac{\ln t}{N(a)}} \right],$$

where $\hat{Q}(a)$ = average reward, $N(a)$ = number of trials of action a .

- As $N(a)$ grows, the bonus shrinks $\Rightarrow$ converges to exploitation.
- Actions with **fewer trials** get more chances $\Rightarrow$ principled exploration.

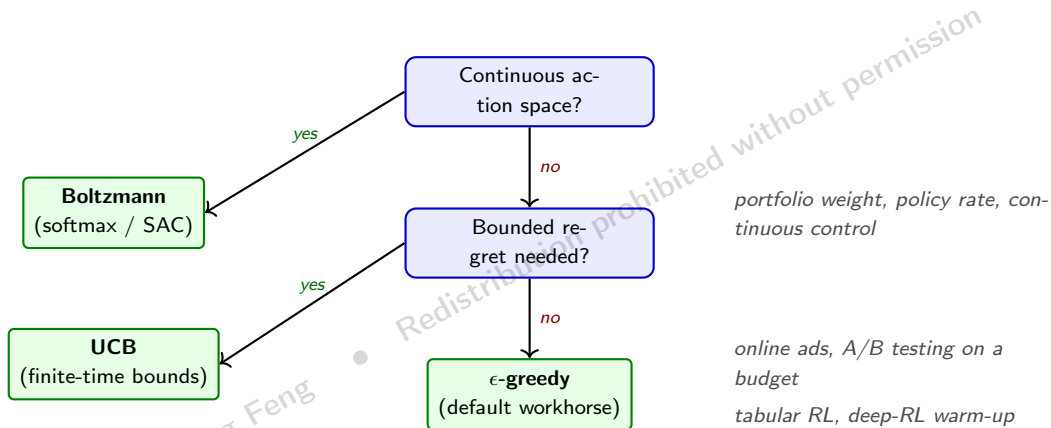
Economic example: A firm evaluating new markets. Early data is sparse, so a confidence bonus encourages trying each region enough to reduce uncertainty.

Comparison Table — ϵ -Greedy / Boltzmann / UCB

Method	Hyperparameter	Exploration mechanism	Sample complexity / when best
ϵ -Greedy	$\epsilon \in [0, 1]$	Uniform random with prob. ϵ ; greedy with prob. $1 - \epsilon$	$\tilde{O}(\mathcal{A} /\epsilon^2)$ visits; default workhorse for tabular and deep RL
Boltzmann	Temperature $\tau > 0$	Softmax over Q , $\Pr(a s) \propto \exp(Q/\tau)$	Smooth; favors near-optimal actions; natural for continuous $\mathcal{A}$
UCB	Exploration weight α	Bonus $\alpha\sqrt{\ln t/N(a)}$ on under-trying actions	$\tilde{O}(\sqrt{ \mathcal{A} t \ln t})$ regret; principled but assumes stationarity

Optimistic initialization is a complementary trick: start $Q(s, a)$ large so every action is tried at least once. Often combined with ϵ -greedy in early training.

When to Use Which: Decision Rule by Problem Class



Choice also depends on: cost of a bad action, size of $\mathcal{A}$, whether $Q(s, a)$ -uncertainty can be quantified.

Temporal-Difference Learning and Q-Learning

Update every step: sampling (like MC) + bootstrapping (like DP)

From MC to TD: The Key Innovation

Monte Carlo: waits until the **end of an episode** to update.

- Analogy: estimate a mean by collecting a **full batch** of data, then averaging.

Temporal Difference (TD): updates **after every single step**.

- Analogy: **incremental mean update** after each new data point.

From statistics:

- Non-incremental: $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i \rightarrow$ collect n samples, then compute.
- Incremental: $\hat{\mu}_{k+1} = \hat{\mu}_k + \alpha_{k+1}(X_{k+1} - \hat{\mu}_k) \rightarrow$ update after each sample.

TD applies this incremental logic to **value function estimation**.

TD(0): The Simplest TD Method

MC update:

$$V(s_t) \leftarrow V(s_t) + \alpha \left(\underbrace{G_t}_{\text{full return}} - V(s_t) \right).$$

TD(0) update:

$$V(s_t) \leftarrow V(s_t) + \alpha \left(\underbrace{r_{t+1} + \gamma V(s_{t+1})}_{\text{one-step target}} - V(s_t) \right).$$

Key difference:

- MC uses the **actual full return** G_t (unbiased, high variance).
- TD uses a **bootstrapped target** $r_{t+1} + \gamma V(s_{t+1})$ (biased, lower variance), and learns **online**.

TD Error: $\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t)$ (the “surprise” signal).

Biased, but self-correcting: the target aims at the Bellman image $T^\pi V$, not V^π . While V is wrong it is biased — but T^π is a contraction, so as $V \rightarrow V^\pi$ the bias vanishes.

TD(0) Applied to the Growth Model

Setup: State $s_t = (k_t, z_t)$, policy π chooses c_t , reward $r_{t+1} = u(c_t)$.

Each time step:

1. Observe state (k_t, z_t) , choose c_t via policy π .
2. Receive reward $r_{t+1} = u(c_t)$.
3. Observe next state (k_{t+1}, z_{t+1}) .
4. Update:

$$V(k_t, z_t) \leftarrow V(k_t, z_t) + \alpha \left[u(c_t) + \beta V(k_{t+1}, z_{t+1}) - V(k_t, z_t) \right].$$

Advantage over VFI: no need to sum over all possible z' . Just use the **realized** transition.

Advantage over MC: no need to simulate an entire lifetime. Update **every period**.

MC vs. TD(0) vs. DP: Comparison

Dimension	Monte Carlo	TD(0)	Dynamic Prog.
Bias / variance	Unbiased / high	Biased / low	Exact (no sampling)
Bootstrapping	No (full return)	Yes (1-step target)	Yes (full backup)
Requires model?	No	No	Yes (P , R)
On-/off-policy	Either	Either	N/A (planning)
Sample complexity	Episodes	Steps	Sweeps (no samples)
Convergence	Asymptotic (LLN)	Asymptotic	Finite (contraction)

Key insight: TD sits between MC and DP. It uses **sampling** (like MC) and **bootstrapping** (like DP), combining the best of both worlds.

On-Policy vs. Off-Policy Learning

- **On-Policy (SARSA):**

- The *behavior policy* (collects data) = the *target policy* (being improved).
- Updates based on the **action actually taken**:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)].$$

- **Off-Policy (Q-Learning):**

- The behavior policy can **differ** from the target policy.
- Updates assume **greedy** next action, regardless of what was actually done:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)].$$

Economic insight: Off-policy = learn about the **optimal** strategy while exploring with a different behavior. Like studying the best possible portfolio allocation while actually holding a diversified “safe” portfolio.

Comparing SARSA and Q-Learning

Feature	SARSA (On-Policy)	Q-Learning (Off-Policy)
Update target	$r + \gamma Q(s', a')$	$r + \gamma \max_{a'} Q(s', a')$
Behavior = target?	Yes	No (can differ)
Converges to	Q^π (of <i>actual</i> behavior)	Q^* (optimal)
Risk awareness	Reflects real actions	Plans for “ideal” actions

Which to choose?

- **SARSA**: when **safety** matters — learning matches actual behavior (e.g., autonomous driving).
- **Q-Learning**: when you want the **optimal** policy directly and can tolerate off-policy mismatch.

Off-Policy Evaluation: Learning a New Rule from Old Data

Goal: evaluate a *target* policy π using trajectories generated by a *different behavior* policy μ (e.g. historical data).

$$\mathbb{E}_{\pi}[G_t] = \mathbb{E}_{\mu}[\rho_{t:T-1} G_t], \quad \rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(a_k | s_k)}{\mu(a_k | s_k)}$$

the **importance-sampling** (reweighting) ratio.

Volcker as a “behavior policy.” Pre-1979 vs. post-1979 Fed behavior $\approx$ two different Taylor rules (Clarida–Galí–Gertler 2000). IS lets us estimate the value of a *new* candidate rule π from historical data — **without re-running the policy experiment**. This is *counterfactual policy evaluation*.

Two cautions. (i) **Coverage / overlap:** need $\mu(a|s) > 0$ wherever $\pi(a|s) > 0$ — the RL version of the *common-support* condition in causal inference (the ratio is dynamic IPW). (ii) The *product* of ratios can make the variance **explode** over long horizons.

Q-Learning in the Growth Model

Watkins & Dayan (1992), *Machine Learning*, "Q-learning"

- **Action-value function:** $Q(k, c) = u(c) + \beta \mathbb{E}[V(k') \mid k, c]$.
- **Optimal policy:** $\pi^*(k) = \arg \max_c Q^*(k, c)$.
- **Q-Learning update (online, model-free):**

$$Q_{m+1}(k_t, c_t) = Q_m(k_t, c_t) + \alpha \left[u(c_t) + \beta \max_{c'} Q_m(k_{t+1}, c') - Q_m(k_t, c_t) \right].$$

- **Advantages:**
 - Works with stochastic or **unknown** transitions (sample-based).
 - Can approximate $Q(\cdot)$ with neural networks for continuous state/action spaces.
 - Converges to Q^* under standard conditions.

Multi-Step Methods: TD(λ)

Idea: Combine partial returns over n steps with bootstrapping.

- $n = 1$: recovers TD(0).
- $n = T - t$: recovers MC.
- Intermediate n : balances bias and variance.

TD(λ) — eligibility traces:

- Blends all n -step returns using exponential weighting parameter $\lambda \in [0, 1]$.
- $\lambda = 0$: pure TD(0). $\lambda = 1$: pure MC.
- Often improves convergence in practice.

One dial, two endpoints: λ slides continuously between **TD(0)** ($\lambda=0$: one-step, low variance) and **Monte Carlo** ($\lambda=1$: full return, unbiased). Every method in Act 5.2 is a choice along this single bias–variance dial.

Bridge: From Tables to Function Approximation

Act 5.2 delivered the model-free core. MC, TD(0), SARSA, and Q-learning all learn values and policies from sampled experience — tied together by one backbone, Generalized Policy Iteration.

- But every method so far stores one number per state (or state–action pair) — a **table**. That is exactly the wall Act 5.1 warned about: economic states (capital, prices, wealth distributions) are **continuous and high-dimensional**, so the table explodes.
- **Next (5.3):** replace the table with a **neural network**. Function approximation turns tabular Q-learning into **deep RL** (DQN); the policy-gradient / **actor–critic** family then handles continuous actions — and we see where the recipe has actually worked (AlphaGo, RLHF, AlphaManager, Krusell–Smith).

Arc: 5.1 MODEL (framework + DP) → 5.2 SAMPLE (MC / TD / Q-learning) → 5.3 SCALE (actor–critic + deep RL + applications).